Lecture-8
Fluid Kinematics III:
Flow lines, streamfunction, and irrotationality

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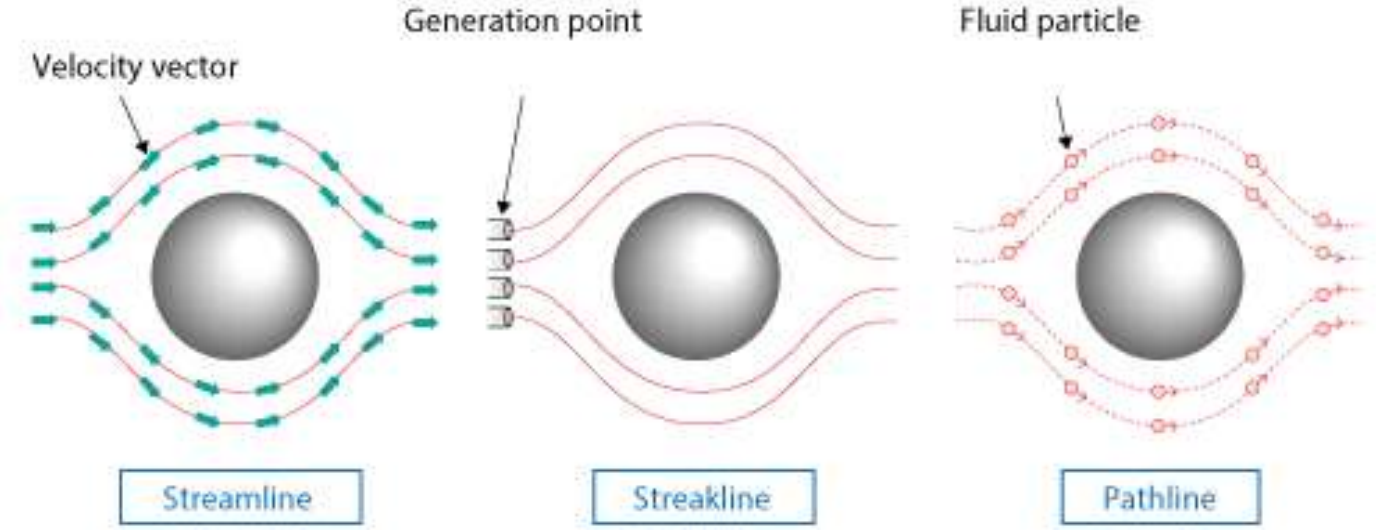
Learning Objectives

By the end of this lecture, you should be able to

1. Distinguish between **streamlines**, **streaklines**, and **pathlines**
2. Formulate the **equation for a streamline** given the velocity field
3. Derive the **2D streamfunction** from the incompressibility constraint
4. Show that the value of the **streamfunction is constant along a streamline**

Flow visualization lines

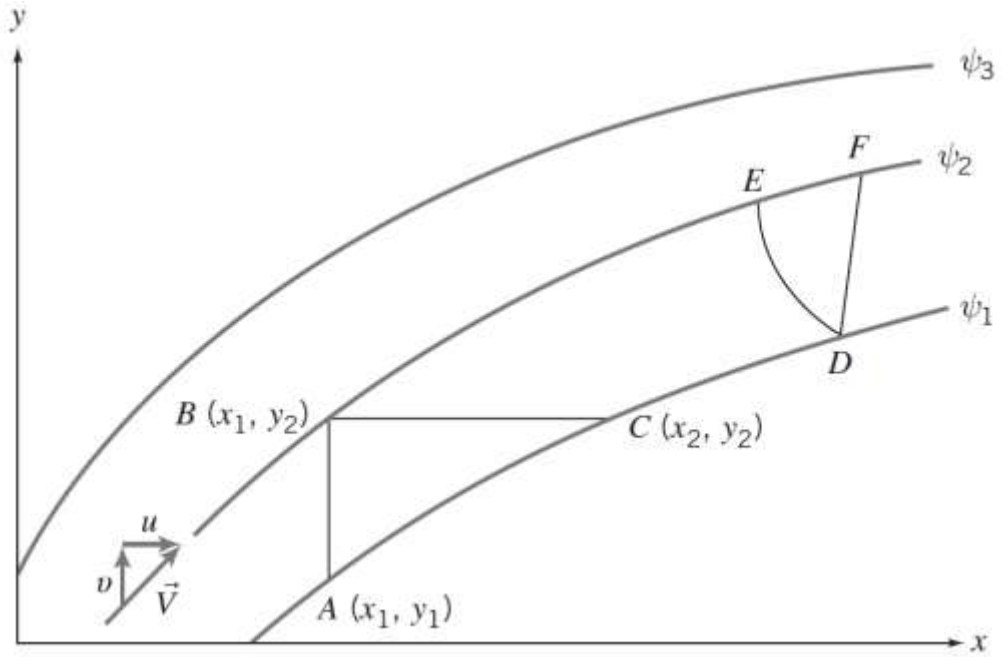
- **Streamlines** are lines drawn in the flow field so that at a given instant, they are **tangent to the direction of flow** at every point in the flow field.
- **Pathline** is the path or trajectory traced out by a **moving fluid particle**.
- If we **identify a location in space** and **trace all the fluid particles** that have passed through that location at some point in time, then the **line joining these fluid particles** is referred to as the **streakline**.



https://www.cradle-cfd.com/media/column/a147

Streamfunction for a 2D incompressible flow

- Recall from the previous lecture that the **volume dilation rate** of a fluid is given by the **divergence of velocity**.
- The **volume flow rate per unit depth** (because we are working in 2D) between two streamlines is the **difference between the two streamfunction values**.



Fox, Robert W., Alan T. McDonald, and John W. Mitchell. 2020. *Fox and McDonald's Introduction to Fluid Mechanics*. 10th ed. Hoboken, NJ: John Wiley & Sons.

Revisiting the learning objectives

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